Chapter 5: Non-ideal Interconnect Issues

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1.1 5.1 Why Losses Matter *(p.3)*

As digital systems push to higher speeds: - Smaller geometries + higher signal rates **frequency content rises** - Resistive losses become more severe - **Signal amplitude decreases** reduced noise margins and slower edge rates timing degradation

Two types of **dispersion** arise: 1. Loss varies with frequency 2. Phase velocity varies with frequency Both are caused by the non-ideal characteristics of real transmission lines.

1.2 5.2 DC Conductor Losses *(p.4)*

DC resistance depends on: - **Resistivity** of the conductor material - **Cross-sectional area** of the current path

$$R_{\text{DC}} = \frac{\rho \cdot L}{A}$$

(chap5.pdf, p.4)

- Primarily a concern for very small geometry conductors, very long lines, and multiload buses.
- Ground return path losses are usually **negligible** at DC (large cross-sectional area).

Note: For FR4 substrates, when frequency increases beyond DC up to 15 GHz, the **substrate (dielectric) loss dominates** over the conductor loss. *(p.5)*

1.3 5.3 Skin Effect *(pp.512)*

1.3.1 Definition *(p.6)*

At high frequencies, current **migrates toward the periphery** (skin) of the conductor. The skin depth is defined as the depth at which the electric field amplitude decays to e^{-1} of its surface value:

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}} = \sqrt{\frac{\rho}{\pi F \mu}} \quad \text{meters}$$

(chap5.pdf, p.6)

where ρ = resistivity (ům), μ = permeability (H/m), F = frequency (Hz).

~63% of the total current flows within one skin depth.

As F increases, δ decreases effective cross-section shrinks **resistance increases as** $\sqrt{F}$.

1.3.2 Frequency-Dependent Inductance *(p.7)*

At DC, current fills the conductor uniformly (including the interior) **internal inductance** adds to external inductance.

At high frequencies, current is confined to the skin only **external inductance** remains. The total inductance decreases asymptotically to the external inductance value.

For most high-speed digital applications, the frequency is high enough that internal inductance can be **ignored**.

1.3.3 Conductor Loss in a Microstrip *(pp.812)*

Current concentrates on the **bottom edge** of the signal line (pulled toward the ground plane by the E-field). Also curves up the sides due to the thickness t.

Approximate total AC resistance (microstrip):

$$R_{\text{ac, microstrip}} = \sqrt{\rho\pi\mu F}\left(\frac{1}{W} + \frac{1}{6H}\right) \quad \Omega/\text{m}$$

(chap5.pdf, p.12)

(Signal line contribution $\approx \sqrt{\rho\pi\mu F}/W$; ground plane contribution $\approx \sqrt{\rho\pi\mu F}/(6H)$, assuming 79.5% of return current flows within $\pm 3H$.)

More exact formula (conformal mapping):

$$R_{\text{signal}} = [\text{loss ratio}]\left(\frac{1}{\pi} + \frac{1}{\pi^2}\ln\frac{4\pi W}{t}\right)\frac{\sqrt{\pi\mu F\rho}}{W}$$

where Loss ratio $= \begin{cases} 0.94 + 0.132\frac{W}{H} - 0.0062\left(\frac{W}{H}\right)^2 & 0.5 < W/H < 10 \\ 1 & W/H \leq 0.5 \end{cases}$

(chap5.pdf, p.12)

Stripline: Current is distributed between top and bottom conductor surfaces. AC resistance parallel combination of top and bottom portions. *(p.13)*

1.3.4 Surface Roughness *(pp.1415)*

Real metal surfaces have a **tooth structure** roughness increases effective resistance by 1050% when mean roughness skin depth.

- Typical FR4 boards: average tooth size = **47 m**
- Etched surfaces: smaller tooth size

Surface resistance:

$$R_{\text{ac}} = R_S\sqrt{F}$$

(chap5.pdf, p.15)

where $R_S = \sqrt{\rho\pi\mu}$ in units of ůs/square (a material constant, independent of frequency and geometry). The full ac resistance per meter of line is:

$$R_{\text{ac signal}} = R_S\sqrt{F}\cdot\frac{L}{W} \quad \Omega/\text{m}$$

(chap5.pdf, p.16)

1.3.5 Effect of AC Losses on Signal Waveforms *(pp.1719)*

A periodic square wave at 50% duty cycle has Fourier components at the fundamental and odd harmonics only:

$$v(t) = \frac{4A}{\pi}\left[\sin(\omega_0 t) + \frac{1}{3}\sin(3\omega_0 t) + \frac{1}{5}\sin(5\omega_0 t) + \cdots\right]$$

(chap5.pdf, p.17)

In a matched system, the **attenuation factor** α can be calculated from the voltage divider at a single frequency using VNA measurements.

Visual effect on the waveform *(p.19)*: - High-frequency components are attenuated more waveform **rounding** (slower edges) - Low-frequency amplitude also decreases overall **amplitude reduction**

1.4 5.4 Frequency-Dependent Dielectric Losses *(pp.2027)*

1.4.1 Physical Mechanism *(p.20)*

A time-varying electric field causes **polar molecules** in the dielectric to align with the field. The damping/relaxation mechanisms dissipate energy and vary with frequency.

1.4.2 Complex Dielectric Constant *(p.20)*

$$\varepsilon = \varepsilon' - j\varepsilon''$$

The imaginary part ε'' represents losses.

1.4.3 Loss Tangent *(p.21)*

$$\tan|\delta_d| = \frac{1}{2\rho\pi F\varepsilon} = \frac{\varepsilon''}{\varepsilon'}$$

(chap5.pdf, p.21)

This is equivalent to a conductivity $1/\rho = 2\pi F\varepsilon''$.

1.4.4 Shunt Conductance in TX Line Model *(p.21)*

The dielectric loss appears as a shunt conductance per unit length in the RLCG model:

$$G = \frac{\varepsilon''}{\varepsilon'}(2\pi F C_{11}) \quad \text{Siemens/meter}$$

(chap5.pdf, p.21)

where C_{11} is the self-capacitance per unit length.

1.4.5 FR4 Loss Tangent *(p.22)*

FR4 has $\tan\delta \approx 0.02$ (given in chap4; varies moderately with frequency). The glass cloth reinforcement contributes negligible dielectric variation with frequency in typical ranges.

1.4.6 Calculating Losses Example *(pp.2327)*

Given: $W = 5$ mil, $H_1 = H_2 = 10$ mil, $t = 0.63$ mil, $\varepsilon_r = 4.0$, stripline.

Steps: 1. Calculate R_S (surface resistivity) 2. Find the frequency where skin depth = conductor thickness ($\delta = t$); below this, use DC resistance; above this, $R \propto \sqrt{F}$ 3. At 400 MHz: compute R_{ac} from the skin-effect formula 4. Compute G at 400 MHz from $G = \tan\delta \cdot 2\pi F C_{11}$ (using PD to find C_{11})

(chap5.pdf, pp.2327)

1.5 5.5 Variations in the Dielectric Constant *(pp.2830)*

1.5.1 Why ε_r Varies *(p.28)*

Properties that depend on ε_r: propagation velocity, characteristic impedance, and crosstalk.

ε_r is NOT constant it varies with: - Frequency - Temperature - Moisture absorption - Spatial variation in composite material proportions

1.5.2 FR4 Composition *(p.29)*

FR4 = epoxy matrix + woven glass cloth. The dielectric properties vary substantially sample-to-sample.

First-order approximation:

$$\varepsilon_{r,\text{FR4}} \approx V_{\text{gls}}\varepsilon_{r,\text{gls}} + (1 - V_{\text{gls}})\varepsilon_{r,\text{epoxy}}$$

(chap5.pdf, p.29)

where V_{gls} = volume fraction of glass.

The glass cloth experiences **no dielectric constant variation** with frequency; the epoxy (resin) variation dominates.

1.6 5.6 Serpentine Traces *(pp.3133)*

1.6.1 Why Used *(p.31)*

Length equalization of traces is required by timing specifications, but real-estate constraints force the use of serpentine (meandering) traces.

1.6.2 Problem *(p.3132)*

The parallel sections of the serpentine **couple** to each other. Due to this coupling, **part of the signal arrives earlier** than it would on a straight trace **ledges** appear on the waveform.

- Duration of the ledges physical length of the coupled sections (L_P)
- Magnitude of ledges depends on spacing between parallel sections

1.6.3 Rules of Thumb *(p.33)*

1. Minimum spacing between parallel sections $S \geq 3H$ to $4H$ (H = height above ground).
2. Minimize L_P (length of serpentine sections).
3. Embedded microstrips and striplines are better (less effect).
4. **Do not serpentine clock traces.**

1.7 5.7 Intersymbol Interference (ISI) *(pp.3438)*

1.7.1 Definition *(p.34)*

ISI occurs when reflections, crosstalk, or other noise has **not fully settled** when the next bit is launched. The unsettled residue corrupts the new symbol.

Most severe when: **period < 2 » TD** (the signal period is smaller than the two-way propagation delay).

1.7.2 Simulation *(pp.3536)*

Proper ISI analysis requires long pseudo-random bit patterns (PRBS). An efficient shortcut:

Simulate at: - Fastest period: `010101010101010` - 2» period: `001100110011001` - 3» period: `000111000111000`

This captures most ISI effects in much less time than PRBS.

1.7.3 Rules of Thumb *(p.38)*

- Minimize reflections (avoid impedance discontinuities, stub lengths, and large parasitics from packages/sockets/connectors).
- Keep interconnects as short as possible.
- Avoid tightly coupled serpentine traces.
- Avoid line lengths that cause signal integrity issues (ringback, ledges, overshoot) coincident with bus transitions.
- Minimize crosstalk.

1.8 5.8 Effects of 90ř Bends *(pp.3941)*

1.8.1 Capacitive Effect *(p.39)*

A 90ř bend introduces **excess capacitance** due to the corner area:

$$C_{90\text{ bend}} \approx C_{11} \cdot w$$

(chap5.pdf, p.39)

where C_{11} = self-capacitance per unit length, w = trace width. The excess area is approximately 0.054 square (full square basis).

1.8.2 Mitigation *(p.40)*

1. **Rounding** the corner eliminates the effect (but problematic for some layout tools)
2. **Chamfering** 45ř diagonal cut at the corner
3. **Using 45ř bends** instead of 90ř bends altogether

1.8.3 High-Frequency Current Paths *(p.41)*

At high frequencies, some current component **hugs the inner corner** of a 90ř bend signals arrive slightly early at the destination.

1.9 5.9 Effects of Topology *(pp.4244)*

1.9.1 Balanced T-Topology *(p.42)*

When the T-junction has equal stub lengths and equal loads signal integrity is preserved with matched or mismatched loads.

1.9.2 Unbalanced T-Topology *(pp.4344)*

With unequal stub lengths or unequal loading, the signal at each leg sees a different discontinuity significant **waveform distortion** and signal integrity degradation.

Key lessons: 1. **Symmetry** of topology is critical (both stub length and loading of each leg). 2. **Minimize discontinuities** at the topology junction.